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Declining Bill Payment Success rate Jul - Aug 2023

From Data to Insight

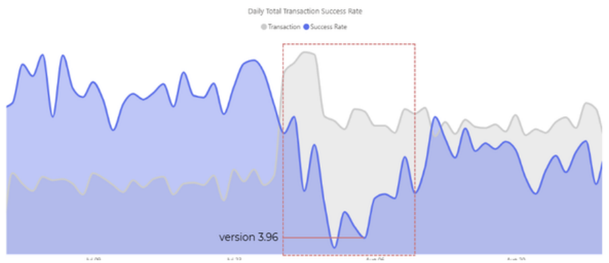
Get Started

Agenda

- Context
- Problem Overview
- User Journey
- Error Analysis
- Segment Analysis
- Recommendations

Root cause 1 — Billing cannot scale after version 3.96

Billing is the largest transaction category and is the main reason for the overall success rate drop.

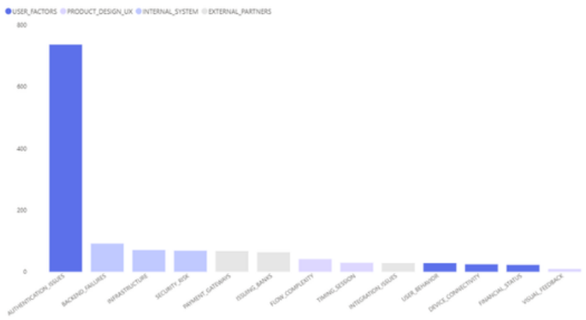


Improve billing scalability

Scale the billing to handle peak transaction volume without degrading success rate.

Root cause 2 — Authentication is the main failure point

This is a flow specific reliability problem, not a platform wide issue

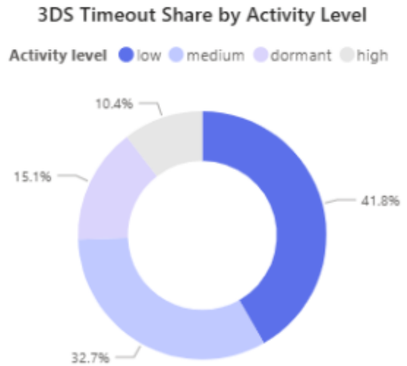


Fix authentication timeout and one time password reliability

Give users enough time to complete authentication by setting the timeout based on real completion data.

Root cause 3 — Failures concentrated in core but less-active users

The timeout setting does not match real user behavior.



Establish lifecycle based performance monitoring

Monitor lifecycle cohorts to detect authentication issues before they impact revenue.

Business Presentation

Root cause identification – Logic tree approach

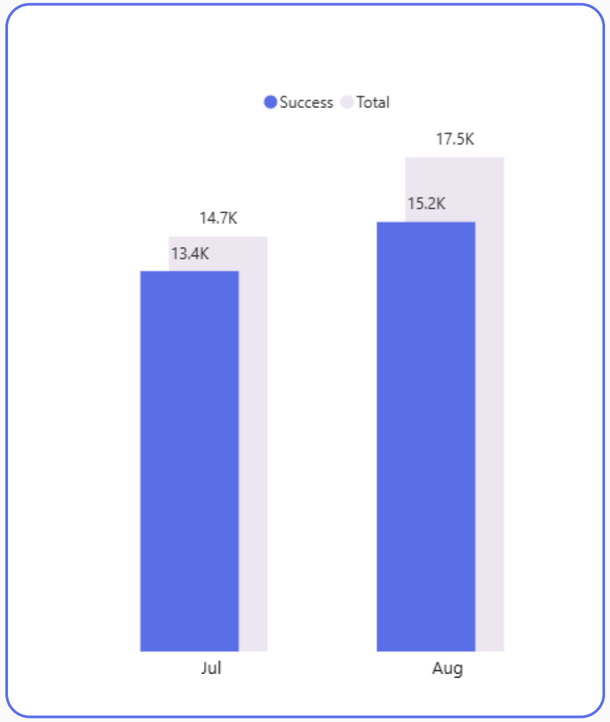
Hypothesis tree combining trend, funnel, error breakdown, and user segment analysis to isolate failure drivers.

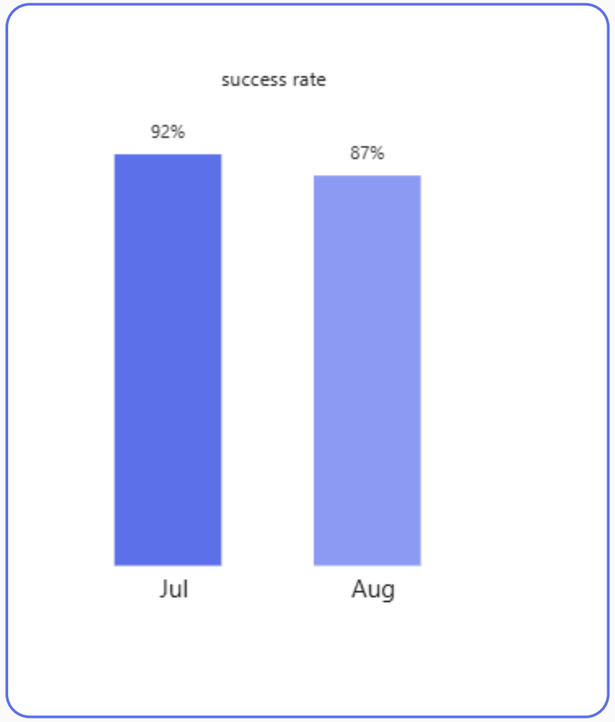
ICE prioritization for key issues

Hypothesis	Impact	Confidence	Ease	ICE Score	Priority
Infrastructure / backend timeout causes drop-off	9	8	8	576	1
UI/UX flow makes users spend too long and hit timeout	8	7	7	392	2
Payment gateway instability during peak time	8	6	6	288	4
Issuer bank rejection rate is higher	7	6	6	252	5
First-time users have lower success rate	6	7	8	336	3
Old devices / OS cause higher failure	5	6	7	210	6
Users with low balance / exceeded limit	4	8	9	288	4

From July to August, transaction volume grew by 23%, but the platform failed to scale with demand, with success rate dropping from 92% to 87%

The opposite movement between volume and success rate indicates a scalability issue: growth exposed underlying system weaknesses and converted a performance gain into a reliability risk.

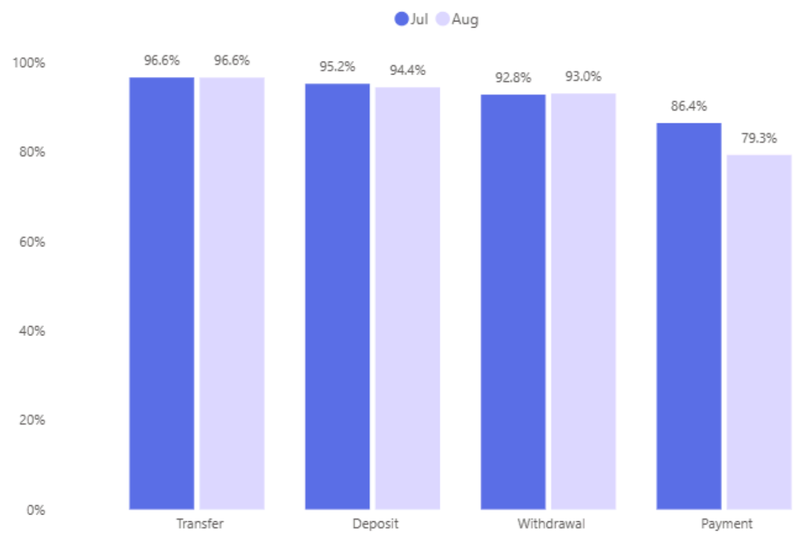




The success rate decline is concentrated in the **Payment** flow (-7.1%), mainly driven by **Billing** - the largest category, while other flows saw only slight declines with minimal impact

The overall decline in success rate is entirely driven by Payment, which dropped by 7.1% in August, while other transaction types remained stable — indicating a product-specific issue rather than a platform-wide problem

The Payment performance drop is mainly driven by Billing — the largest category (37% of volume) and the one with the biggest decline in success rate (-17%), while other categories showed only minor decreases

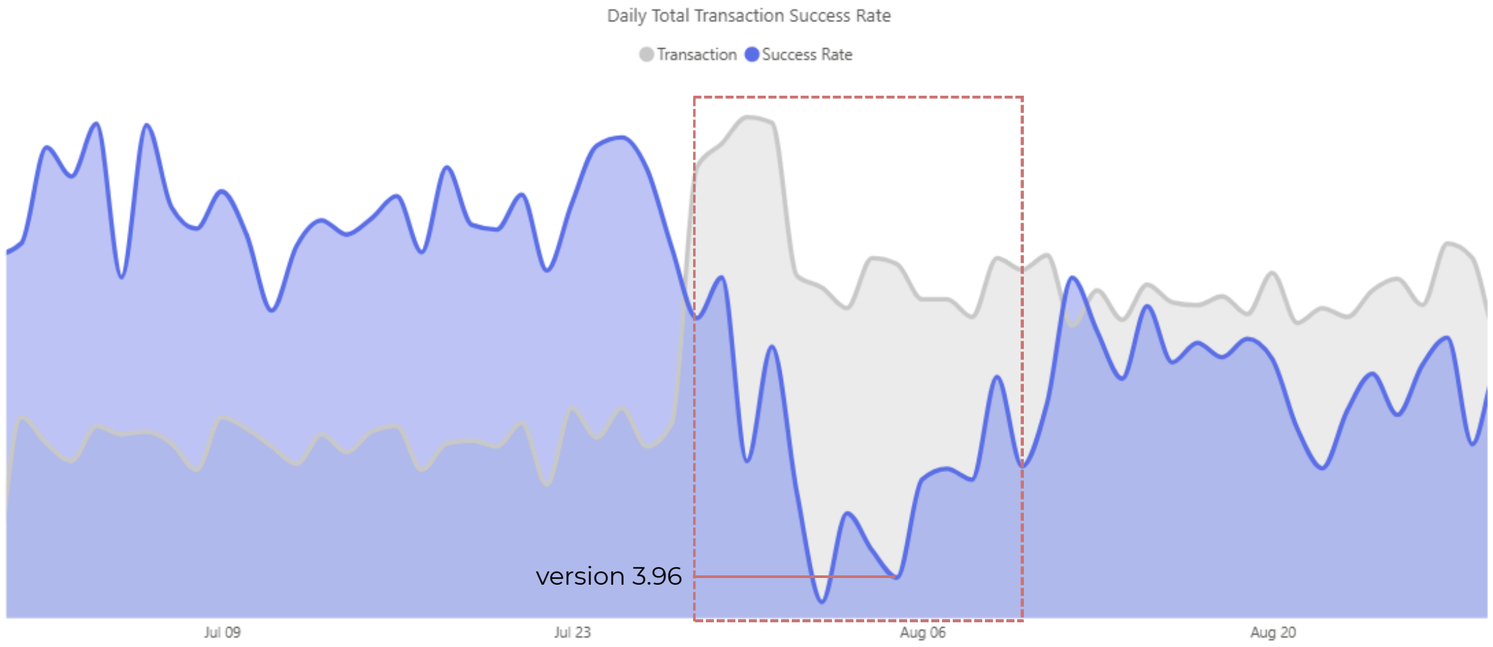


Category	Sucess rate_Jul	Sucess rate_Aug	%Change	%Share
Billing	86%	72%	-17%	37%
Traveling	89%	80%	-10%	3%
Movies	86%	79%	-8%	2%
Telco	84%	79%	-6%	7%
FnB	86%	82%	-5%	11%
Transportation	86%	84%	-2%	2%
Marketplace	87%	85%	-2%	14%
Shopping	87%	86%	-2%	24%

Billing success rate remained stable in July but fell sharply right after the version **3.96 release** in early August, when transaction volume peaked, indicating the new version exposed a scalability bottleneck under high load

Success rate remained relatively stable throughout July, dropped sharply in early August, and only partially recovered toward the end of the month, indicating a systemic issue rather than normal day-to-day fluctuation.

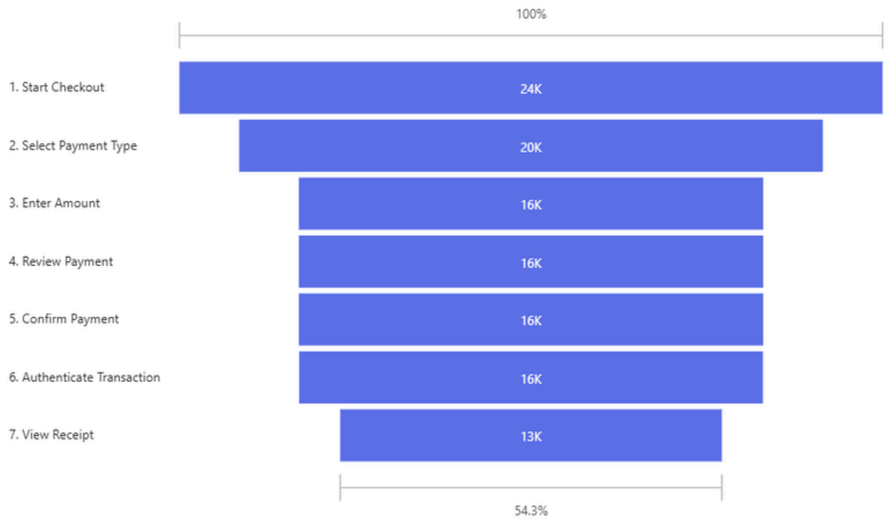
When transaction volume increased in early August, the success rate dropped noticeably, showing that the system could not handle the higher load effectively.



The main bottleneck occurs at the payment setup stage, where users drop off between selecting **the payment type** and **entering the amount**, while the transaction processing and authentication steps show near-perfect completion rates.

The main friction happens before users enter payment details, with ~34% of users dropping during payment setup, while the core transaction steps show near-perfect completion.

After amount entry, completion stays ~100% through confirmation and authentication. Despite being the longest step (~31.6s), authentication keeps a 99.9% conversion. The drop to 82.3% at receipt (-2.7K users) likely reflects users leaving after success, not a transaction issue.

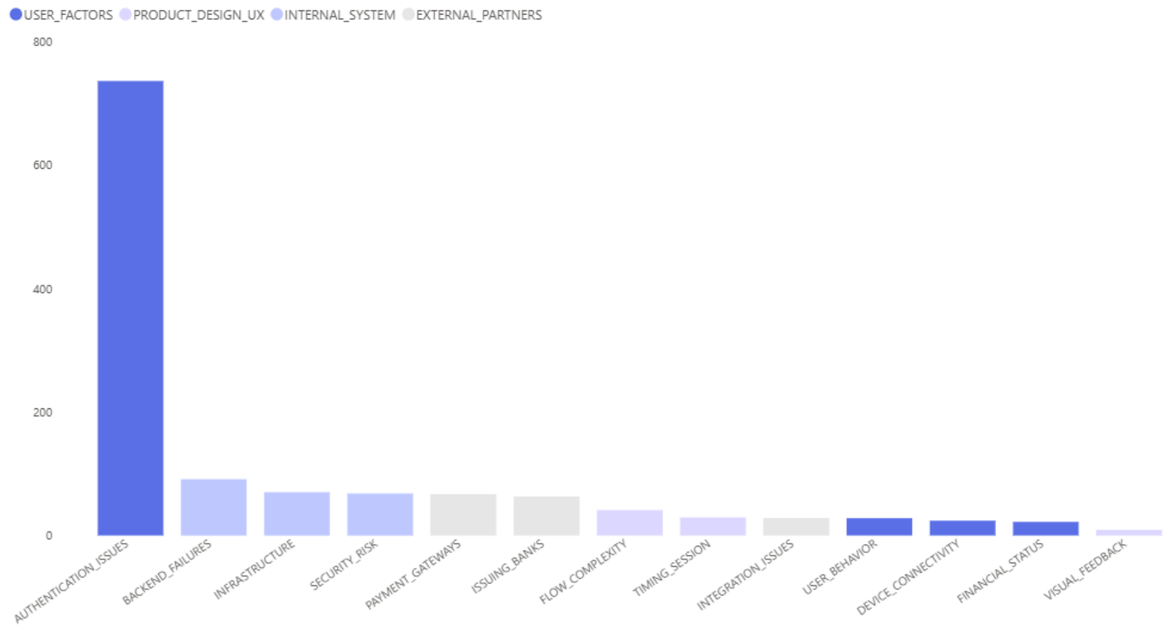


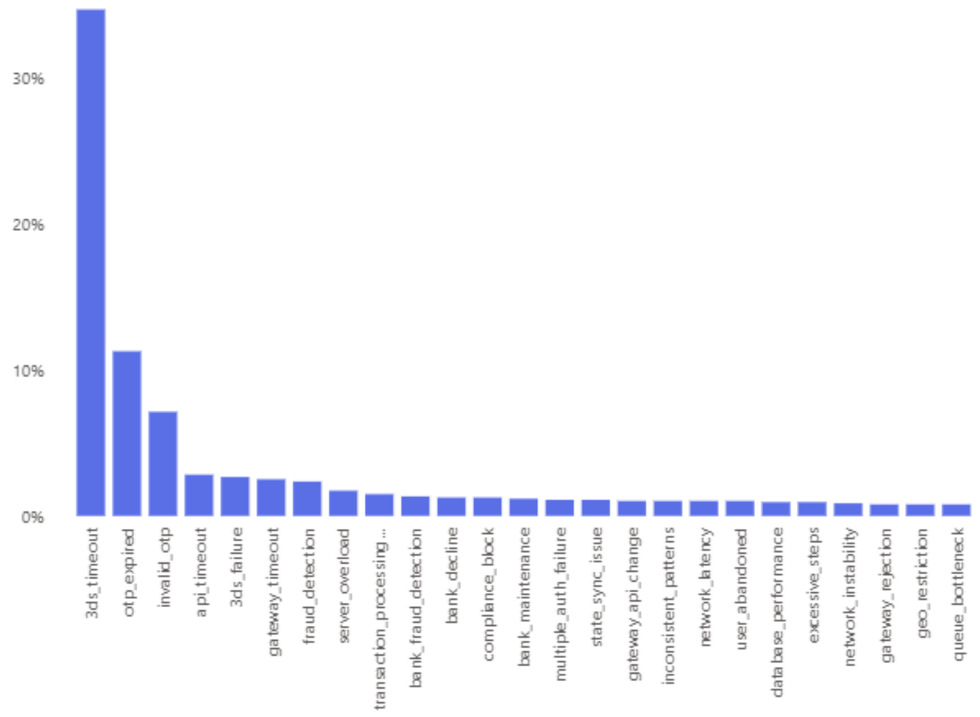
StageName	Count Stage	Percent of First	Percent of Previous	Average Time From First Step	Average Time From Previous Step
1. Start Checkout	23551	100.0%		0.00	
2. Select Payment Type	19551	83.0%	83.0%	5.49	5.49
3. Enter Amount	15551	66.0%	79.5%	28.80	23.28
4. Review Payment	15551	66.0%	100.0%	42.69	13.98
5. Confirm Payment	15551	66.0%	100.0%	71.61	28.79
6. Authenticate Transaction	15531	65.9%	99.9%	97.12	31.63
7. View Receipt	12784	54.3%	82.3%	141.53	27.68

Billing failures are primarily driven by **authentication**, with **3ds timeout** as the single largest root cause, followed by otp_expired and invalid_otp

Authentication issues overwhelmingly dominate all other error groups, far exceeding backend, infrastructure, or external partner failures.

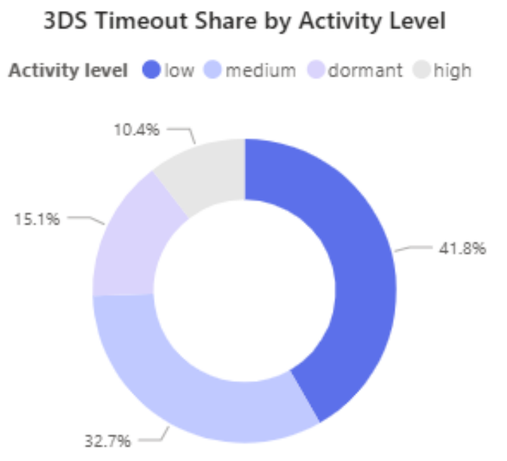
3DS timeout is the dominant failure driver, accounting for the largest share by a wide margin, followed by OTP-related issues, while all other error types contribute only marginally.

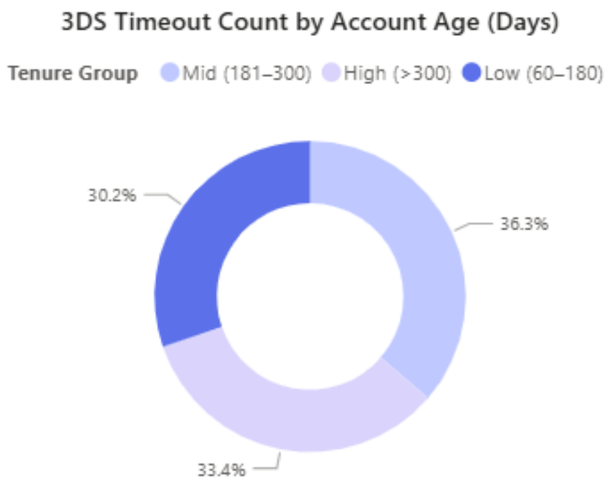


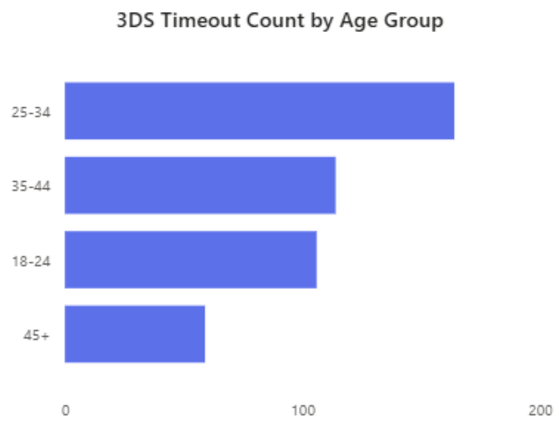


3DS timeout primarily impacts returning but infrequent core users, not new users or power users, indicating the **authentication timeout is too aggressive** for users who are familiar with the product but interact more slowly

Activity level: Errors are concentrated in low and medium activity users, who generate most transaction volume, while high-activity users show more stable authentication performance
Age group: The 25–34 segment contributes the most errors, aligning with the core transacting user base rather than indicating higher failure propensity
Age days: Mid-tenure users (181–300 days) account for the largest share of failures, suggesting the issue peaks at the most active lifecycle stag







01.

Improve billing scalability

Run load and stress testing based on peak traffic.
Optimize payment orchestration and external authentication throughput.
Enable automatic scaling during high demand periods.

02.

Fix authentication timeout and one time password reliability

Set the authentication timeout based on real user completion time.
Monitor one time password delivery and trigger automatic resend when delayed.
Provide real time authentication status to prevent user drop off.

03.

Establish lifecycle based performance monitoring

Track success rate by account age and activity level.
Use mid tenure users as the primary early warning indicator.
Continuously measure real user authentication completion time.

Thank You For Your Attention

I look forward to receiving your feedback to help improve this project. Thank you.

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